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SUPPLEMENTARY INFORMATION

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# Diversity dynamics of Phanerozoic terrestrial tetrapods at the local-community scale

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# Supplementary Information

## Supplementary Results

Linear models of non-flying tetrapod local richness

Across the nine numerically-defined sets of superlocalities, all models apart from the intercept-only were statistically significant (Supplementary Fig. 6; Supplementary Table 1). Akaike weights indicate strongest support for the "species ~ phase" model (median Akaike weight of 0.68, ranging from 0.051–0.74; median adjusted R-squared of 0.679, ranging from 0.511–0.766), with this being the preferred model in five of the nine superlocality sets (especially when using higher richness quantile thresholds). However, there is also some support for the model in which time is a covariate ("species ~ time + phase"; median Akaike weight of 0.276, ranging from 0.232–0.712; median adjusted R-squared of 0.679, ranging from 0.519–0.762). This was the preferred model for the remaining four superlocality sets, especially when using lower richness quantiles to define superlocality sets (using lower richness quantiles may capture localities that are less clearly exceptional). In cases where the favored model was "species ~ time + phase", the coefficient of time was significant with slope of nearly zero (~-0.001).

The intercept-only model and models including time and phase with an interaction term or time alone received little support. This indicates that there is little evidence for substantial increases in local richness within phases, but that the fit of the model is greatly improved by allowing a different intercept in each diversification phase. Of particular importance, slopes for individual regressions applied to data points within diversification phases are non-significant for Pennsylvanian–Triassic and Maastrichtian–Cenozoic, but are significant at $p < 0.05$ for the Jurassic–Cretaceous (Supplementary Table 1).

How sampling intensity influences maximum local richness estimates

*Simulated null distributions*. The simulations show that 1) per-bin maximum local richness will appear to increase through the Phanerozoic even if the underlying distribution remains constant (i.e., there is no change in local richness through time or space; Fig. 2); 2) richness curves based on the 90$^{th}$ percentile (a quantile of 0.9) are relatively robust to this sampling-opportunity bias (Fig. 2); and 3) a simulated null distribution using two underlying local richness pools (divided at the K/Pg boundary) fit the empirical data substantially better than a single Phanerozoic pool, and nearly as well as null distributions using larger numbers of underlying pools (Fig. 2; 3). This supports our interpretation of relative stasis in local richness prior to the K/Pg, followed by an abrupt two- to three-fold increase and continued stasis through the Cenozoic.

When simulating a null distribution using a single Phanerozoic pool of local richness values, maximum local richness appears to rise continuously and steeply through the Paleozoic, less steeply but erratically through the Mesozoic, and flattens out through the Cenozoic (Fig. 2). The empirical curve of per-bin maximum local richness lies largely below the lower confidence interval of the null distribution until the Cenozoic (Fig. 2). The null distribution for the 90$^{th}$ percentile (Fig. 2) is much flatter than for maximum local richness, although confidence intervals are considerably wider prior to the mid-Permian. The empirical curve for richness at the 90$^{th}$ percentile falls below the null prior to the K/Pg, but above the null during the Cenozoic.

In contrast, simulating a null distribution using two pools of local richness, separated at the K/Pg boundary, substantially improves the fit of the empirical data to the simulation, based on the sum of the squared residuals (Fig. 3). Although increases for per-bin maximum

local richness still spuriously appear to be continuous, there is a close correspondance between empirical and simulated richness curves at the 90$^{th}$ percentile. Until the mid-Permian, confidence intervals for richness at the 90$^{th}$ percentile are large for both empirical and simulated distributions. Adding a third underlying pool (either using era boundaries or diversification phases hypothesised on the basis of the empirical data) does not noticeably improve the fit of the empirical and simulated curves (Fig. 2). This suggests that the empirical data is most parsimoniously described by a two-phase pre- and post-K/Pg model of local richness, with relative stasis before and after. The artefactual increases or wide confidence intervals in the simulated distribution prior to the Permian suggest that it may be more difficult to infer local richness patterns during the earliest phase of tetrapod diversification, although exceptional localities such as Richard's Spur at least provide minimum bounds on local richness levels during this interval.

Our simulated null distributions show that the statistical support we recover for phase-shift models, based on fitting models to richnesses of exceptional localities, is conservative. This is because increases in sampling opportunities through time make it more likely that gradual increases would be favoured over phase-shift models, even if the underlying distribution of local richness was static.

*Rarefaction curves*. Rarefaction curves of non-flying tetrapod maximum local richness (Supplementary Fig. 5) broadly support our focal results. In particular, they show that 1) when controlling for the number of sampling opportunities, maximum local richness levels for non-flying tetrapods were very similar across the Paleogene, Neogene and Quaternary (rarefaction curves are nearly identical for most subsampling quotas); 2) that we cannot reject the possibility that maximum local richness did not change between the Jurassic and the Cretaceous; and 3) that Triassic, Permian and Carboniferous levels strongly overlap.

*Correlation analyses*. Maximum local richness per equal-length bin correlates strongly with counts of collections (Spearman's $\rho = 0.85$, $p = < 0.01$) and occupied grid-cells (Spearman's $\rho = 0.8$, $p = < 0.01$; Supplementary Figs 8a–b and 9a). However, correlations between the same variables are considerably weaker after they have been detrended using ARIMA models (counts of collections: Spearman's $\rho = 0.48$ and $p = < 0.01$; counts of occupied grid-cells: Spearman's $\rho = 0.14$, $p = 0.39$; Supplementary Figs 8c–d and 9b). This indicates that the raw relationships are driven by long-term trends of increase, and not by direct, short-term linkage. This is consistent with our interpretation of the link between sampling and maximum observed local richness, which is probabilistic but not additive. Nevertheless, poor sampling in some intervals of the Mesozoic (particularly in the Middle Jurassic and "mid"-Cretaceous) is coincident with (and likely related to) gaps in the record of exceptional fossil tetrapod localities, and this prohibits us from tracing a continuous time series of local richness through the Phanerozoic.

## Patterns of local richness in major tetrapod clades

*Non-avian dinosaurs*: Local richness of non-avian dinosaurs was never high in comparison to small-bodied taxa such as mammals or squamates: remarkably, only 14 localities through the entire Mesozoic have yielded more than 10 non-avian dinosaur genera. Maximum dinosaur richness per collection does not exceed five genera or seven species during the Triassic, but increased through the Jurassic, reaching 19 species/genera at the Dry Mesa Quarry collection of the Late Jurassic Morrison Formation, USA (Fig. 5a). With the exception of Careless Creek Quarry from the Judith River Formation (Campanian; 15 species) and Lull 2 Quarry from the Lance Formation (Maastrichtian; 14 species) of the USA, local species richness of non-avian dinosaurs during the Late Cretaceous is notably lower than during the Late Jurassic. Patterns for formations are broadly congruent (Supplementary Fig. 2). Dinosaur richness increases gradually through the Triassic and Early Jurassic before reaching an apparent peak in the Late Jurassic (~62 species from the unusually vast and stratigraphically-thick Morrison Formation), and another in the Late Cretaceous (~51 species from the Dinosaur Park Formation in the Campanian; ~36 species from the Hell Creek and ~23 from the Lance in the Maastrichtian).

*Non-flying mammals*: Absolute counts of mammal taxa per collection are approximately twice those for non-avian dinosaurs (Fig. 5a and b). Sampled local richness of mammals increases gradually from their origins in the Late Triassic to the Middle Jurassic, before reaching an apparent Mesozoic maximum during the Late Jurassic. Kimmeridgian–Tithonian localities that have been bulk-sampled, such as Guimarota and Quarry 9, Como Bluff, are at least as rich (~33 species) as the richest localities from the Early Cretaceous (~26 species from the Mammal Bed, Durlston Bay (Lulworth Formation, UK; Berriasian) and Maastrichtian (~25 species from Bushy Tailed Blowout and Lull 2 Quarry from the

Lance Formation, and Flat Creek 5 from the Hell Creek Formation; Fig. 5b). However, counts of genera per collection are similar between the Late Jurassic and Maastrichtian, and richness per formation may be higher in the Maastrichtian.

Richness per collection and formation both suggest that mammals on a global level experienced a sharp two-to-three-fold increase in local richness across the K/Pg boundary (Fig. 5b, Supplementary Figs 1c and 2c), consistent with previous findings for North America[1-3]. Maximum local richness subsequently fluctuated around ~60 species per collection throughout the Cenozoic. Even the large increase in the prevalence of cave deposits in the Plio-Pleistocene does not appear to result in a noticeable increase. This is consistent with sampling-standardized analyses of continental-scale diversity in North American land-mammals[4,5]. Excepting the Miocene-aged Norden Bridge Quarry (Valentine Formation, Langhian; 77 species), richness of mammals per collection peaked in the Paleocene–Eocene and dropped during the Oligocene. Counts of mammal species per formation also drop dramatically, halving from the late Paleogene into the Neogene. A mid-Cenozoic decline in local richness of North American mammals was previously reported by Stucky[6]. This decline may be linked to the deterioration of global climatic conditions (particularly in North America, the continent that contributes the strongest signal to the "global" pattern). Global cooling is thought to have driven the Bridgerian Crash in North American mammalian diversity at local and continental scales around 50–47 Ma[7].

*Squamates*: Squamate local richness remained very low (no more than 5–8 species per collection) from their earliest appearance in the fossil record, during the Early–Middle Jurassic, until the Campanian, when counts more than doubled (~19 species from Ukhaa Tolgod, Djadokhta Formation, and ~17 species from Khulsan, Barun Goyot Formation; Fig. 5c; Supplementary Fig. 1d). There is no clear evidence for face-value increases immediately across the K/Pg boundary. In fact, squamates appear to have been substantially affected by

the extinction: during the Paleocene, richness dropped to no more than 11 species per collection. However, by the Ypresian, maximum squamate local richness had once again attained Late Cretaceous levels. These levels (maximally ~20–28 species) broadly persisted until the end of the Cenozoic. Patterns of squamate richness per formation are congruent with counts of taxa per collection, but there is a pronounced peak in richness per formation during the Campanian (Supplementary Fig. 2d and g).

*Pseudosuchians, turtles and lissamphibians*: Ecologically important and abundant but species-poor groups such as pseudosuchians (crocodylians and their stem-group) and turtles show overwhelmingly flat local richness levels throughout much of their evolutionary histories (Supplementary Fig. 18). Pseudosuchian local richness peaked in the Late Triassic, followed by remarkable stasis through the remainder of the Mesozoic. This trend continues through the Cenozoic, until a second peak is reached in the late Miocene. Turtle local richness saw apparent peaks in the Maastrichtian and Ypresian, followed by a slight decline towards the present. However, it is often difficult to diagnose pseudosuchian and turtle species from fragmentary material, and this may depress their sampled richness outside of Lagerstätten-style deposits. Sampled lissamphibian local richness is also low. However, given their high diversity today, this is more likely to be due to low preservation potential or diagnosability. Although very low for most of the Mesozoic, there is an apparent rise in lissamphibian local richness during the well-sampled Maastrichtian. Unlike other major groups, lissamphibian local richness does not increase consistently during the Cenozoic.

*Flying tetrapods*: The diversity dynamics of flying tetrapods in deep time are more difficult to establish. Due to their fragile skeletons, our ability to trace patterns of local richness in birds, bats and pterosaurs depends much more heavily on rare events of exceptional preservation, such as Konservat-Lagerstätten deposits. Nevertheless, a small handful of truly exceptional sites allow us to draw preliminary conclusions about patterns of

local richness in flying tetrapods. We focus on counts of flying taxa per formation, because collections in crucial deposits (such as those from the Early Cretaceous Jehol Group in China, where most fossils are collected by local farmers and precise origins are usually unknown) are geographically over-split, artificially deflating richness estimates. We have selectively updated records for key deposits using counts of species from the literature, because data entry in the PaleoDB for flying tetrapods is currently incomplete.

Our results tentatively suggest that maximum local richness of flying tetrapods increased surprisingly little from the Cretaceous to the Neogene (Supplementary Fig. 15). From the Early Cretaceous of China, 66 flying tetrapod species are known from the Jiufotang Formation (including ~46 birds) and 41 from the Yixian Formation (including ~24 birds), although there may be problems with taxonomic oversplitting in the Mesozoic. By contrast, 77 flying tetrapod species are known from the Lutetian-aged (Paleogene, Cenozoic) Messel deposits, where birds and bats are known from hundreds of virtually complete specimens[8]. Fifty-six named species of bird are known from Messel, and an informal tally including undescribed specimens elevates this number to 70 species[9]. At least 36 species are known from the lower Eocene Green River Formation, USA [including ~34 birds; [10]].

Local richness of flying taxa appears to have increased very little between the early Cenozoic and the Recent. The proportional representation of birds and bats within localities and formations increases spectacularly in the Plio-Pleistocene (Supplementary Fig. 15a and b; 15e–h), caused by dramatic increases in the availability of caves and geologically short-lived depositional contexts (see above; Supplementary Fig. 7). In spite of this taphonomic bias, the spectacular avian local richness at Messel is on par with that of even the richest late Cenozoic deposits: only two Pliocene–Holocene formations exceed 30 species, and only one exceeds Messel in richness (the Pliocene Yorktown Formation, USA, with 87 species). For comparison, eBird (http://www.ebird.org/) suggests present-day avian local richness hotspots

range from 70–130 species. Bat local richness is also highly constrained, being maintained at ~10 species from their initial appearance in the early Eocene until the Recent (Supplementary Fig. 15c) [11,12].

Patterns of non-flying tetrapod richness per formation during the Mesozoic

Were it not for three exceptionally diverse formations, the Morrison Formation in the Late Jurassic and the Hell Creek and Lance formations in the Maastrichtian, we might conclude that maximum richness per formation did not increase from the Triassic to the mid-Cretaceous, before increasing in the Late Cretaceous. Counts of species and genera from the Morrison Formation are at least twice those of any other Jurassic formation. However, the outlier status of the Morrison Formation is likely to result from the fact that it is geographically vast (~1.5 million square km), extending from New Mexico to Montana and encompassing a large variety of paleoenvironments, stratigraphically thick (~50 m), and because it has been intensively sampled for over 100 years[13]. The Hell Creek and Lance formations are 1.5-fold richer than the next richest Cretaceous formations (Djadokhta and Barun Goyot, East Asia; crucially, however, these formations have not been bulk-sampled), and comparable to the Late Jurassic Morrison Formation. We therefore consider it possible that this apparent 1.5-fold increase is the result of more intense sampling during the Maastrichtian, and in particular the increasing importance of isolated mammal teeth obtained by intensive bulk-sampling, as well as more intense taxonomic scrutiny and higher diagnosability of those remains that are collected.

Patterns of local richness within continental regions

Patterns of local richness in Phanerozoic tetrapods are broadly congruent between the patterns derived from the aggregated global data and those for individual continental regions (Supplementary Fig. 17). This is because pooled "global" patterns are strongly driven by North America and Europe, the two continental regions that have historically been best-sampled. North America contributes the most data for many important intervals, including the Permian, Late Cretaceous and Paleogene. All of the key events in Phanerozoic local tetrapod richness are recorded in North America, including the late Carboniferous rise, Permo-Triassic stasis, Jurassic and Maastrichtian increases, and Cenozoic stasis. However, gaps are present in the North American record during the Devonian–Mississippian, Late Permian, Middle Jurassic and Early Cretaceous. Europe corroborates the late Carboniferous rise, Jurassic increase and, to some extent, Cenozoic stasis (although the early Paleogene record is poor). Substantial gaps exist in the European record for the Permo-Triassic, "mid"-Late Cretaceous and early Paleogene.

Records of local richness for other continental regions are much less complete. The South American record is poor prior to the Cenozoic, and the Neogene is better-sampled than the Paleogene. Africa and Asia contribute global maxima in local richness across the Permo-Triassic boundary. Asian localities contribute to global maxima in local richness in the Late Cretaceous (e.g. Ukhaa Tolgod, Campanian). However, aside from one collection in the Early Cretaceous (Ksar Met-Lili, Anoual), African localities do not affect global maxima until the Neogene and Quaternary. With some exceptions (e.g. Riversleigh in Queensland, Australia), geographic regions encompassing equatorial paleolatitudes (including those not contained within our five continental regions; designated in Supplementary Fig. 17 as "Other") do not contribute substantial amounts of data on tetrapod local richness until the Neogene–Quaternary (Supplementary Fig. 16).

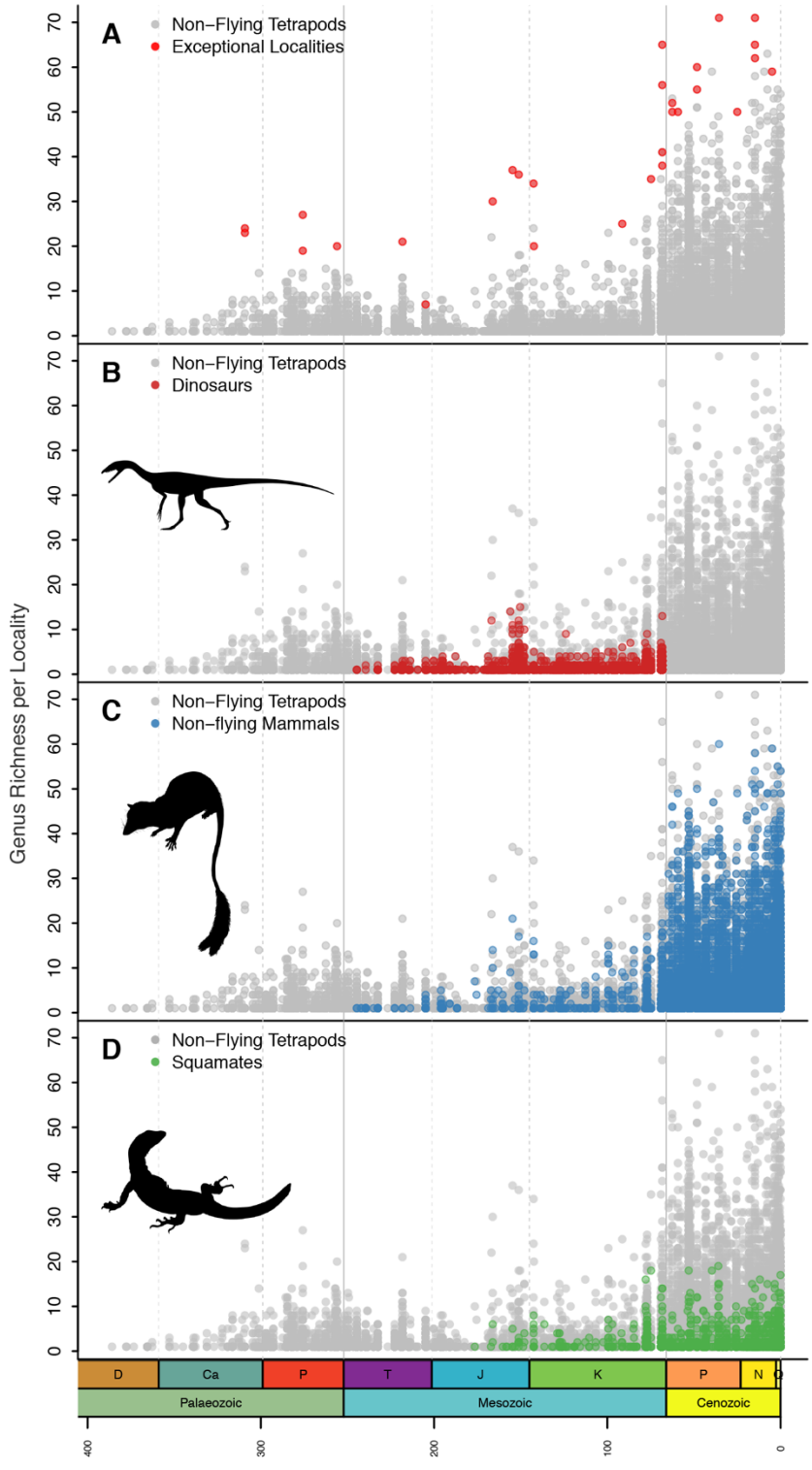
Supplementary Fig. 16 could be interpreted as showing a geographic shift in peak local richness from mid to low paleolatitudes through the latter half of the Cenozoic (perhaps due to the establishment of a modern-style latitudinal diversity gradient). However, this is more likely to be a sampling artefact resulting from the marked increase in the number of localities known from low paleolatitudes in the Neogene and Quaternary (Supplementary Fig. 19).

## Supplementary References

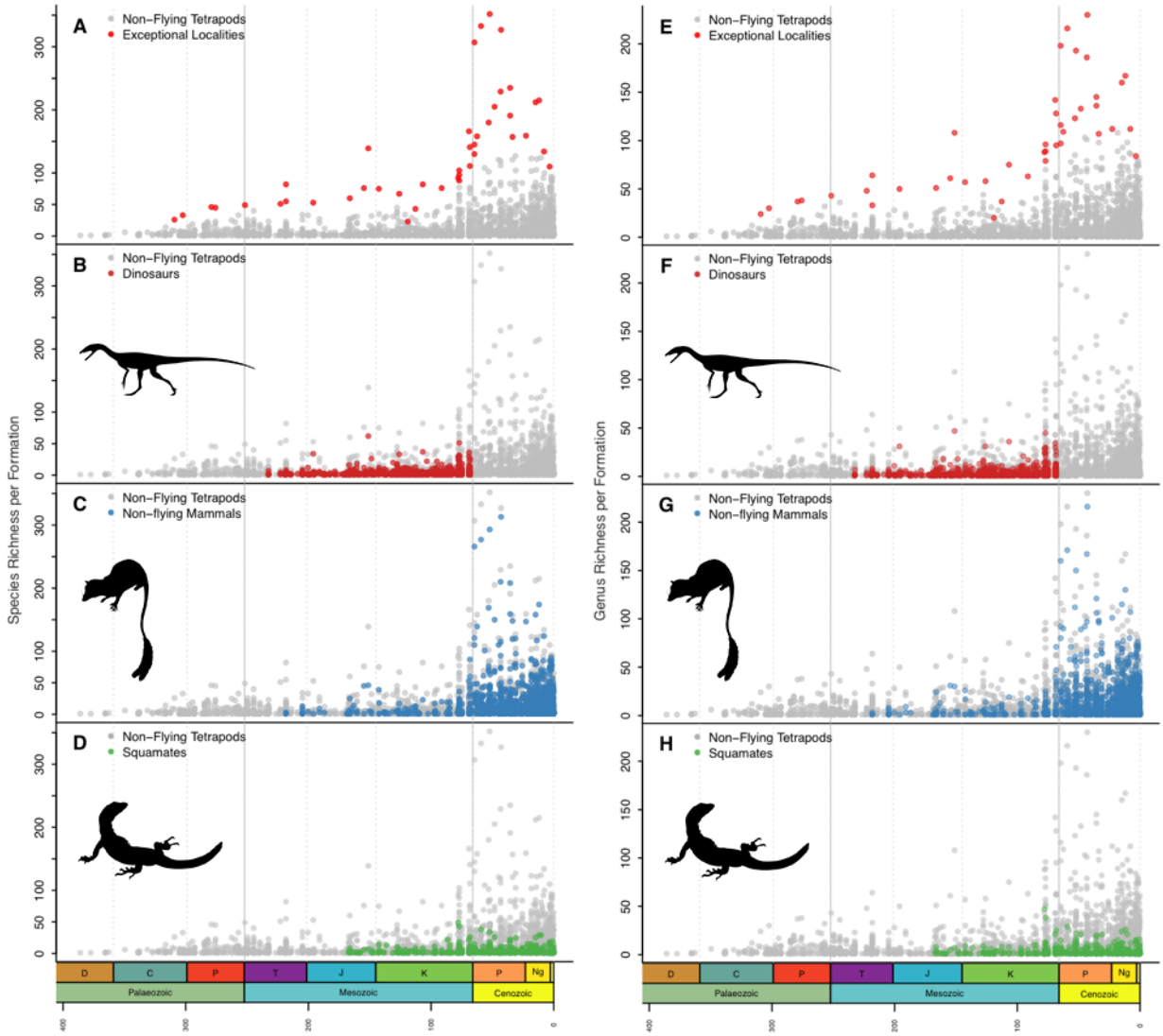

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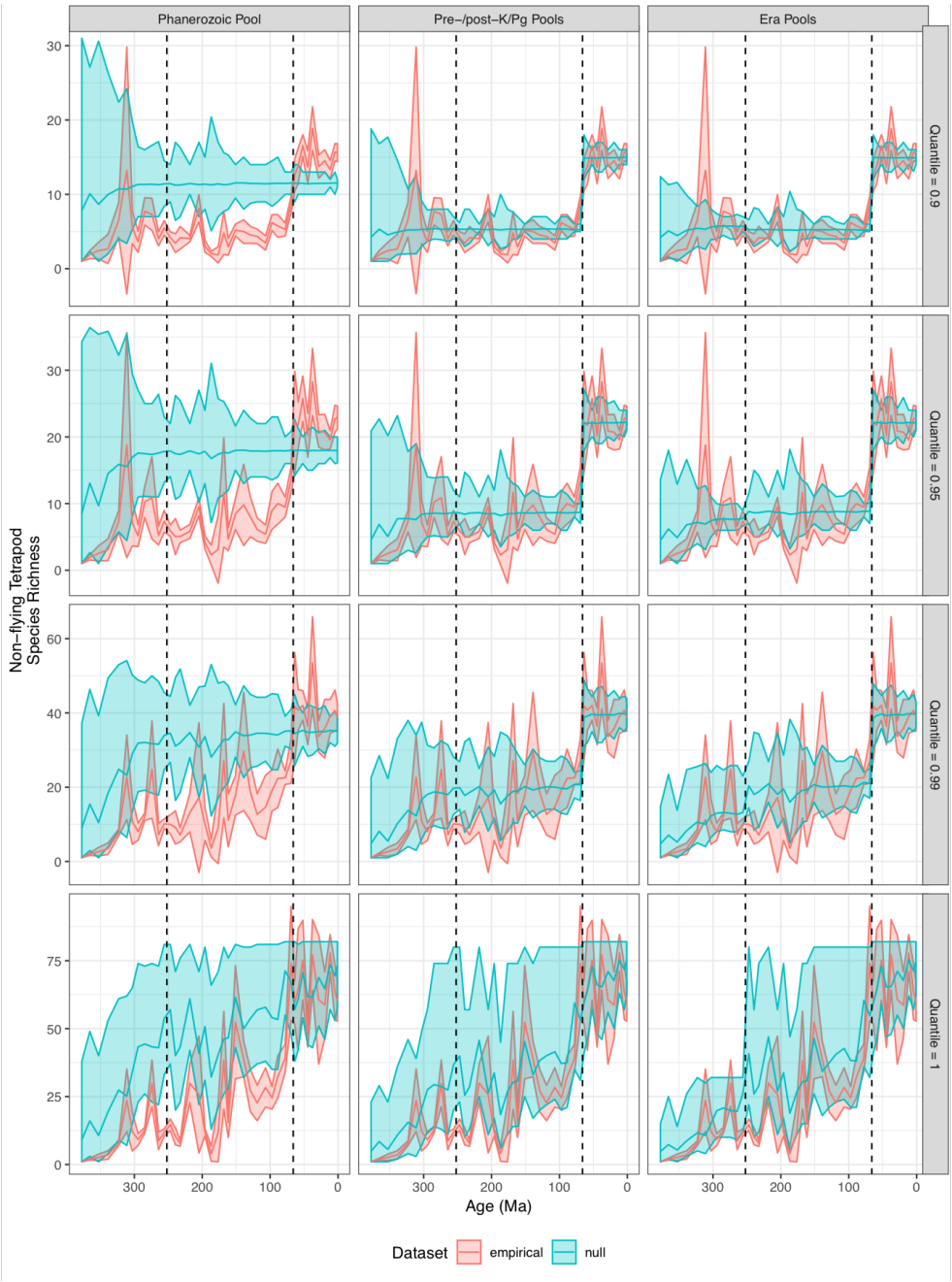
## Supplementary Figures



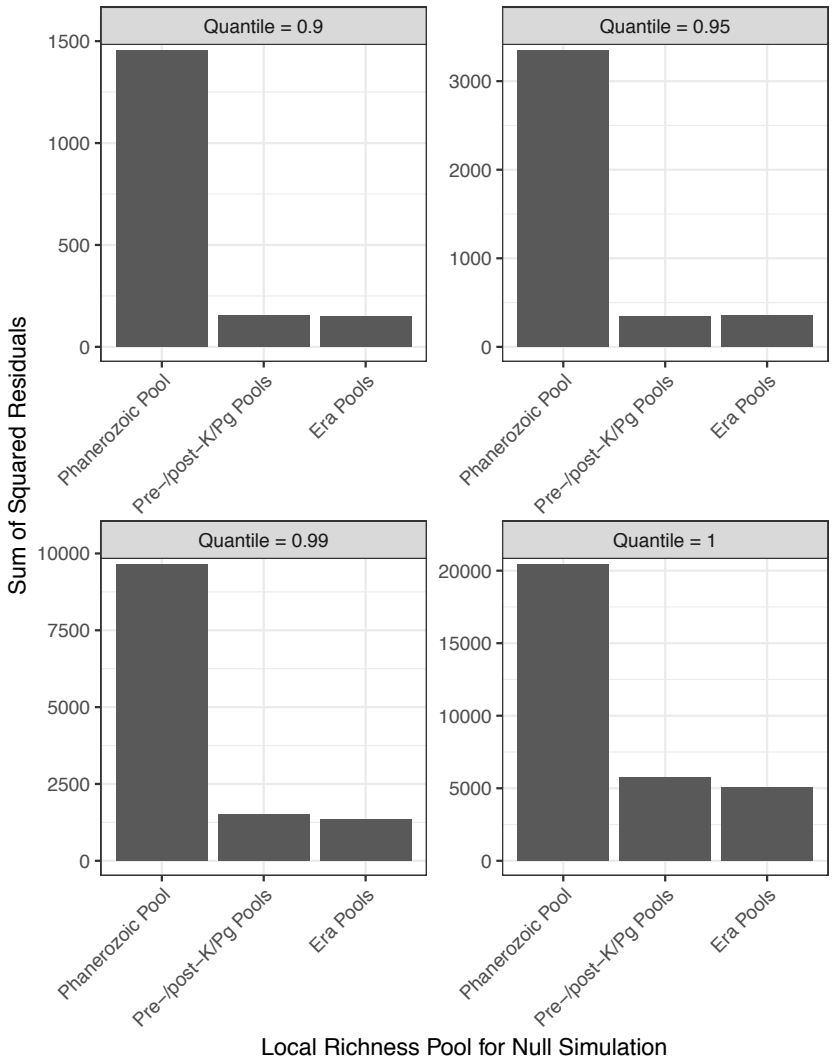

**Supplementary Fig. 1**. Genus-level local richness (taxa per collection) for non-flying, non-marine Phanerozoic tetrapods. **a**, Non-flying, non-marine tetrapods. **b**, Non-avian dinosaurs. **c**, Non-chiropteran mammaliamorphs. **d**, Squamates.



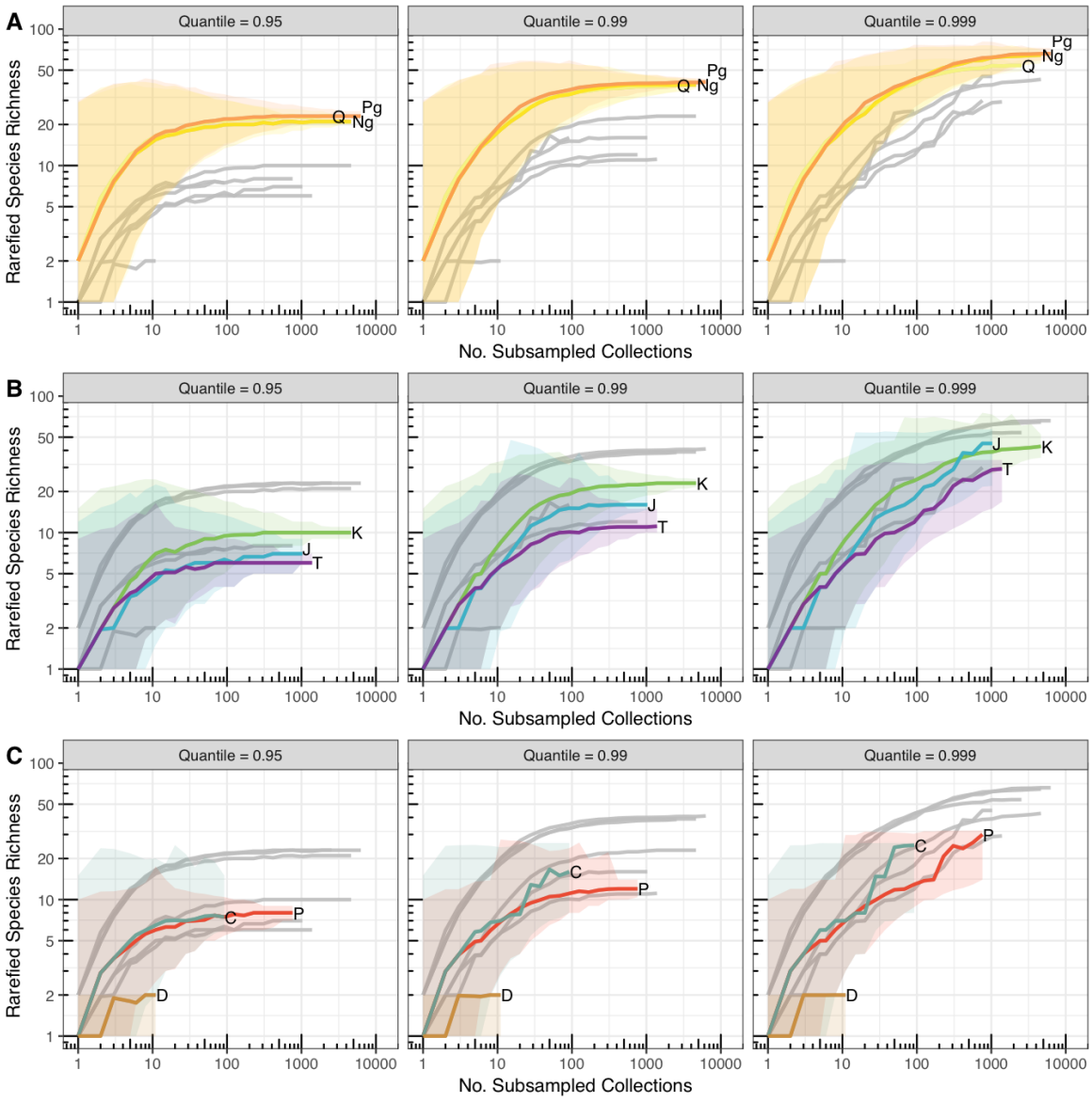

**Supplementary Fig. 2**. Richness per formation for Phanerozoic non-flying terrestrial tetrapods. Species richness for **a**, Non-flying tetrapods. **b**, Non-avian dinosaurs. **c**, Non-chiropteran mammaliamorphs. **d**, Squamates. Genus richness for **e**, Non-flying tetrapods. **f**, Non-avian dinosaurs. **g** Non-chiropteran mammaliamorphs. **h**, Squamates.




**Supplementary Fig. 3.** Comparison between the empirical curve of local species richness (using quantiles of 0.9, 0.95, 0.99 and 1) for terrestrial tetrapods and simulated null distributions. Richness quantiles calculated for equal-length bins.




**Supplementary Fig. 4.** Residual sums of squares for simulated null distributions of local richness (using quantiles of 0.9, 0.95, 0.99 and 1) using Phanerozic, Pre-/Post-K/Pg and Era Pools. Using separate pre-/post-K/Pg pools substantially improves the fit compared to a single Phanerozoic pool, while adding a third pool (Era Pools) makes only a negligible improvement.



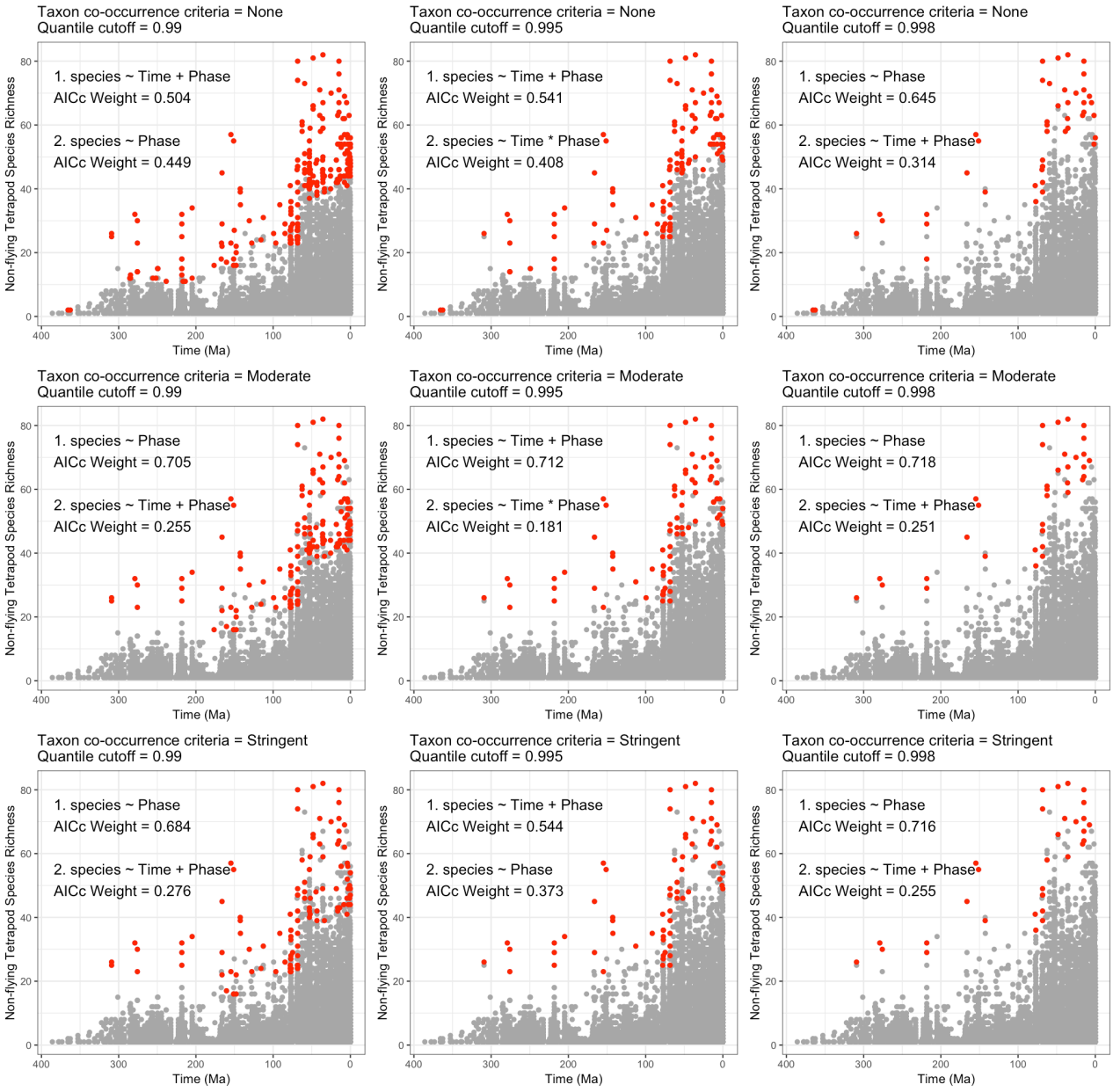

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286 **Supplementary Fig. 5**. **Rarefaction curves for local richness quantiles of terrestrial**

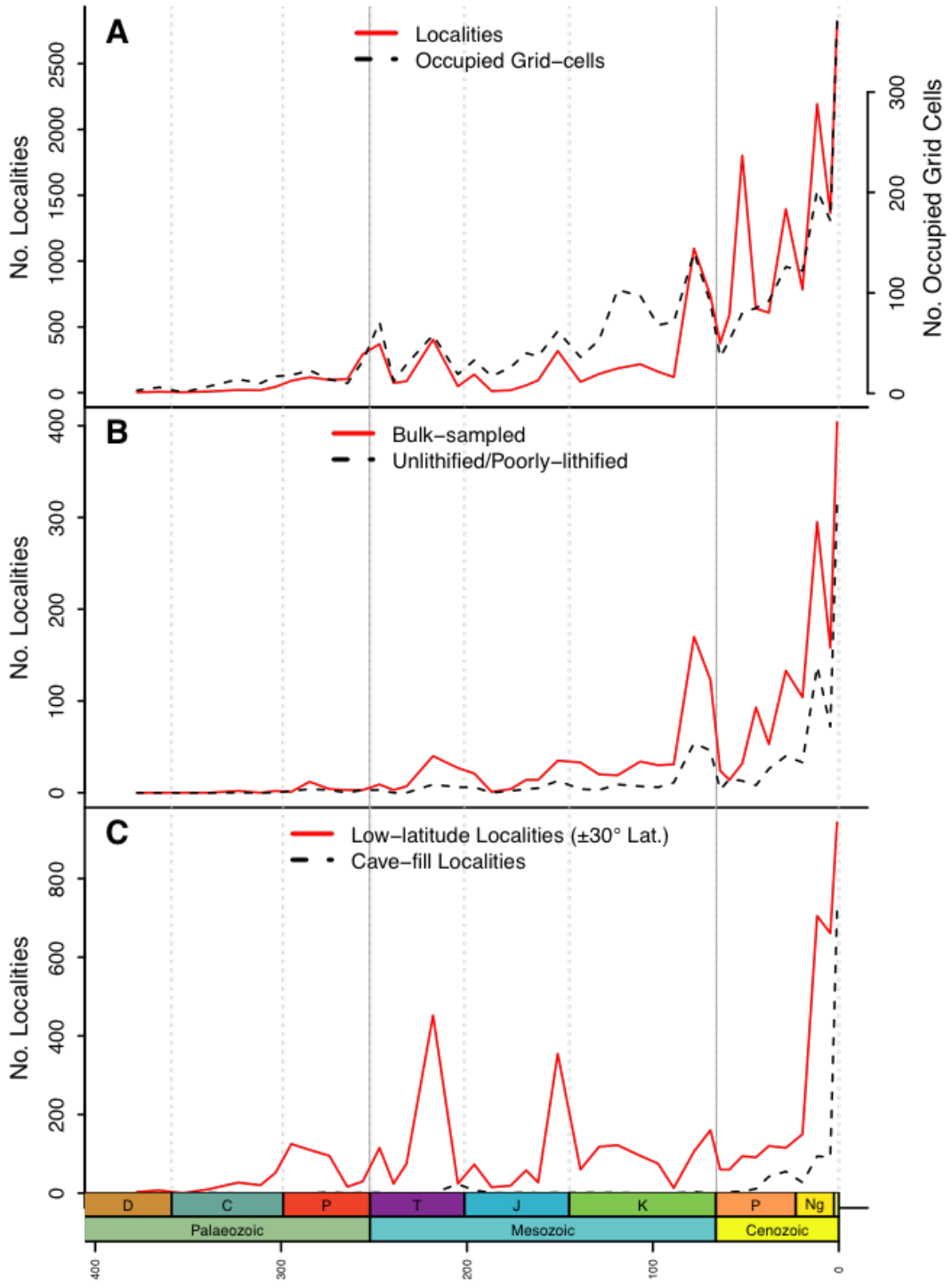
287 **tetrapods, rarefying by collection.** Shaded regions show 95% confidence intervals

288 calculated by bootstrapping (1000 replicates). **a**, Cenozoic. **b**, Mesozoic. **c**, Palaeozoic.

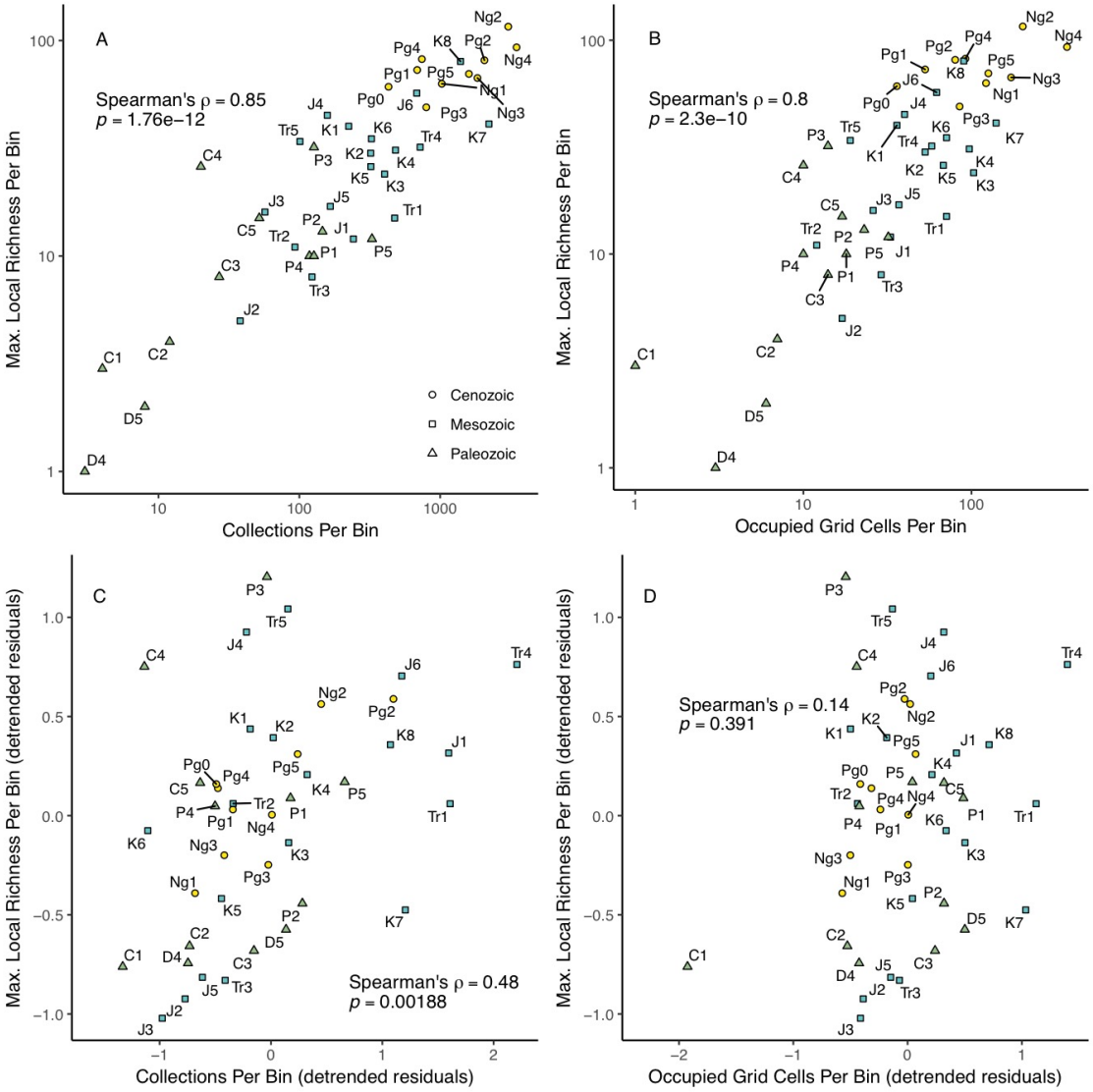
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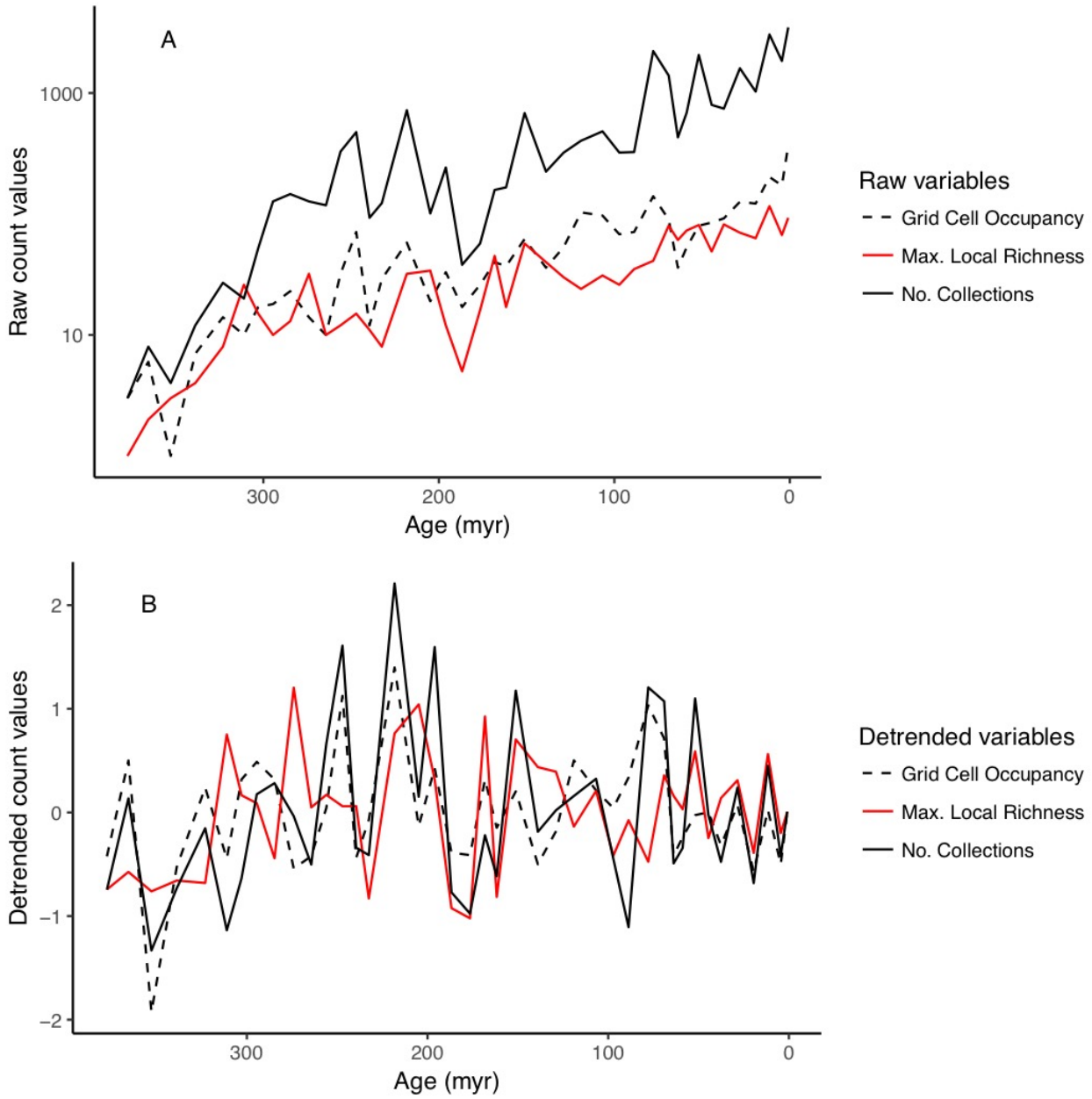

**Supplementary Fig. 6**. Localities selected using numerical definitions of exceptional sites (combining taxonomic co-occurrence criteria with quantile thresholds). See Methods for details.




**Supplementary Fig. 7**. Sampling trends in Phanerozoic tetrapods. **a**, Counts of collections and occupied equal-area grid cells (50 km spacing). **b**, Counts of bulk-sampled and unlithified or poorly-lithified collections. **c**, Counts of low-palaeolatitude collections (within ±30° of palaeolatitude either side of the palaeoequator) and cave-fill collections.

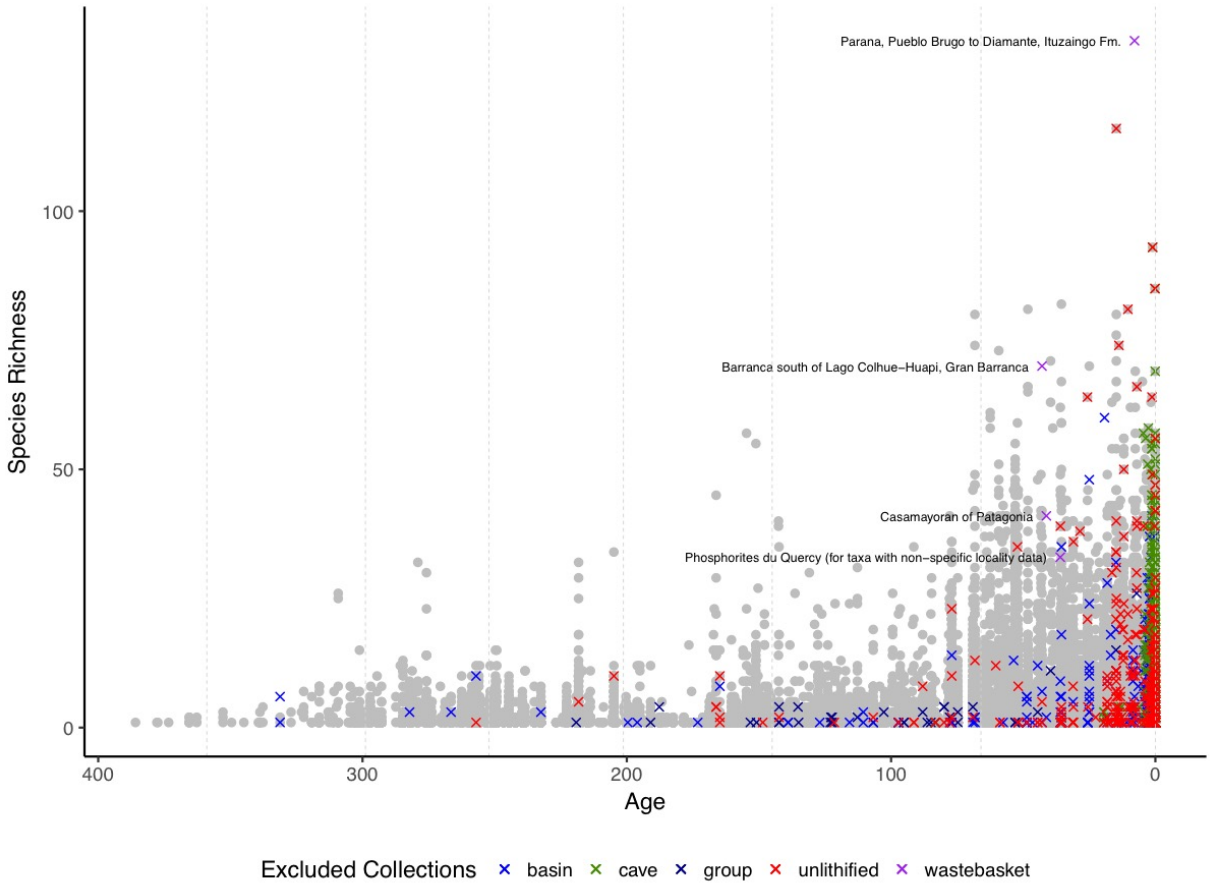



**Supplementary Fig. 8**. Correlation between maximum local richness of non-flying tetrapods and sampling proxies. Points represent equal-length bins. **a**, Raw relationship between maximum local richness of non-flying tetrapods and counts of collections. **b**, Raw relationship between maximum local richness of non-flying tetrapods and counts of occupied equal-area grid cells with 50 km spacings. **c**, Relationship between maximum local richness of non-flying tetrapods and counts of collections after applying first-differencing. **d**, Relationship between maximum local richness of non-flying tetrapods and counts of occupied equal-area grid cells with 50 km spacings after applying first-differencing. First differencing aggressively removes the influence of long-term correlations, so that only short-term correlations remain.




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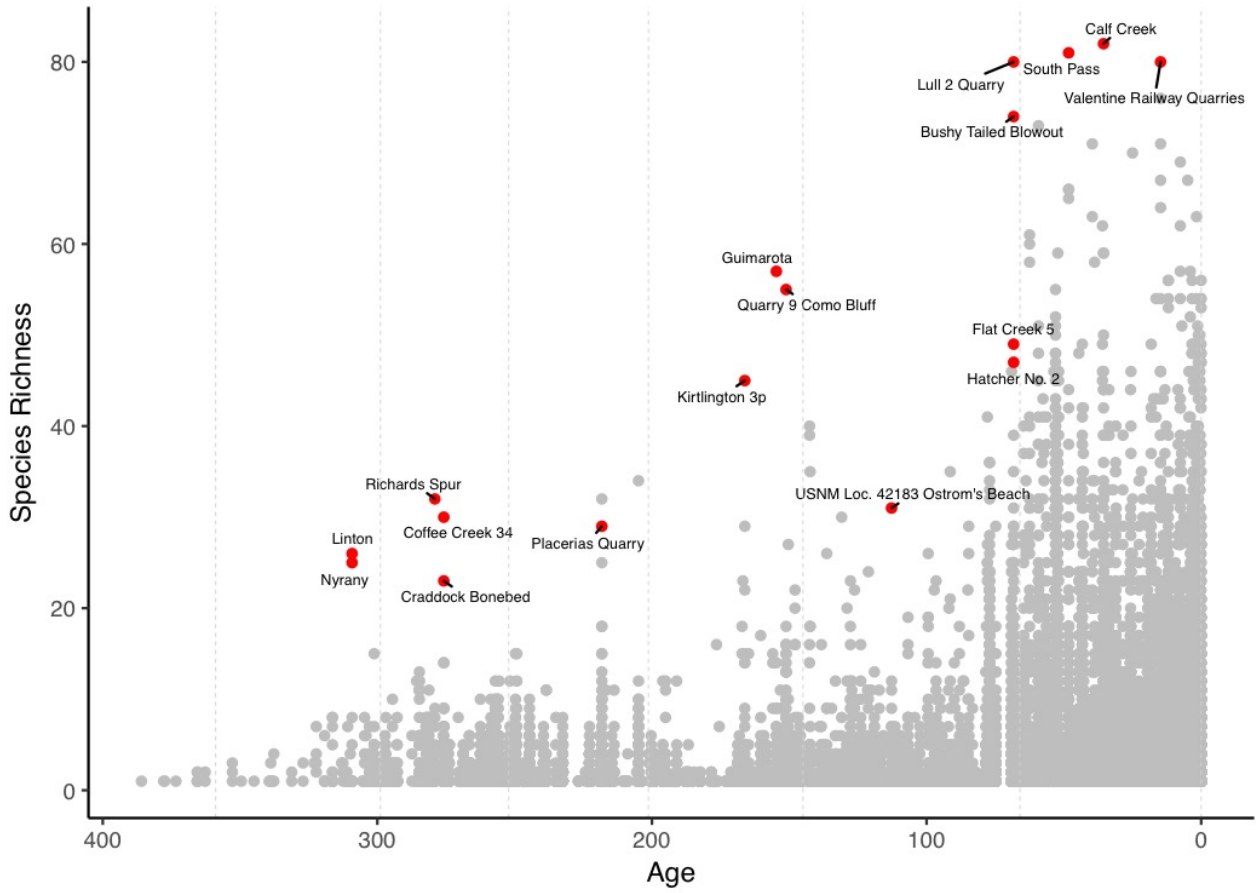
310 **Supplementary Fig. 9**. **a**, Raw and **b**, model-detrended time series of sampling proxies and local richness.




311

312 **Supplementary Fig. 10**. Collections that have been dropped from main figures and analyses,
313 either because they represent "wastebasket" collections in the Paleobiology Database;
314 preservation conditions that are difficult to compare with earlier parts of the fossil record
315 (unlithified sediments, cave-fill environments); basin-level geographic scales; or group-level
316 stratigraphic scales.

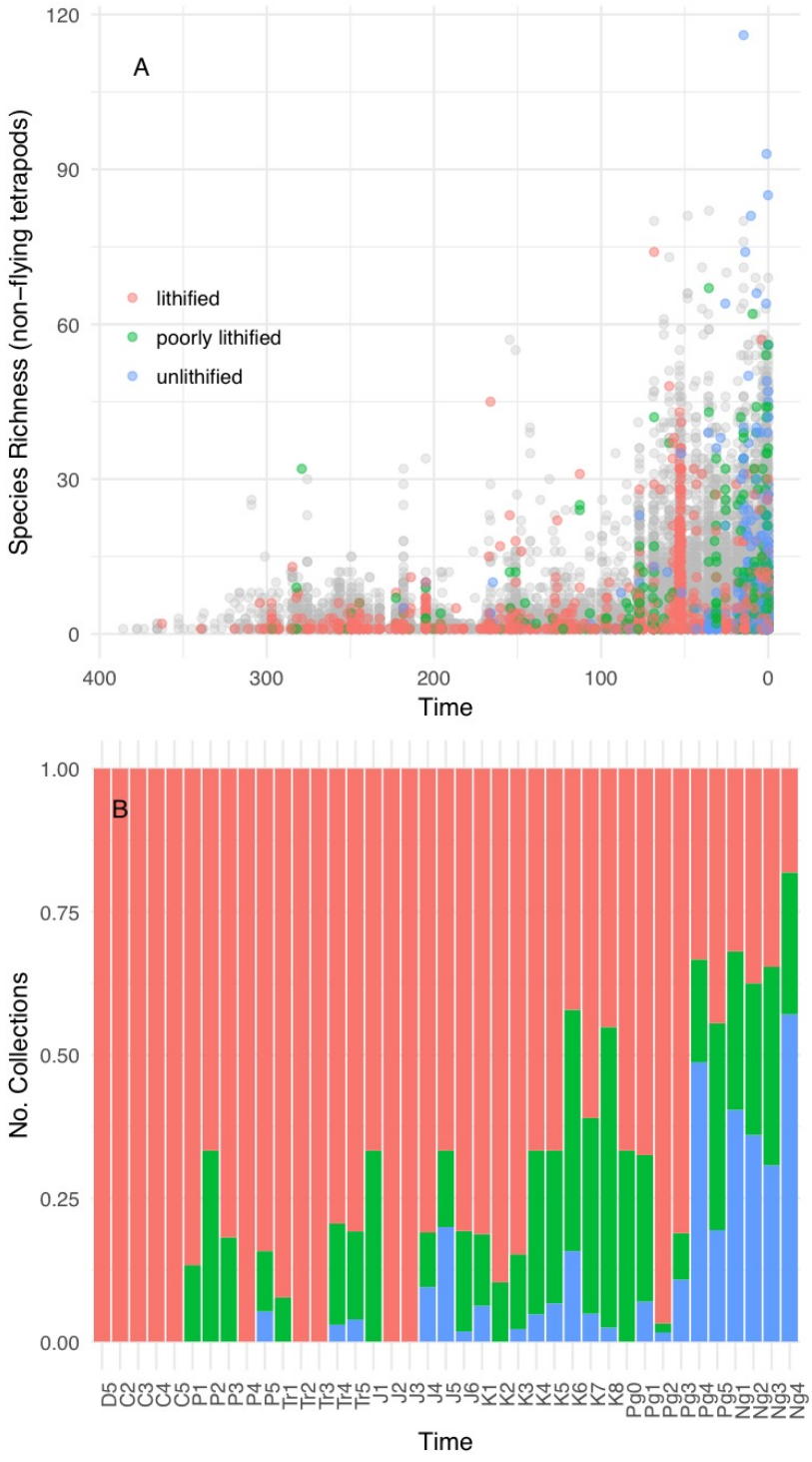
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319 **Supplementary Fig. 11**. Identities of selected exceptional localities through the Phanerozoic.

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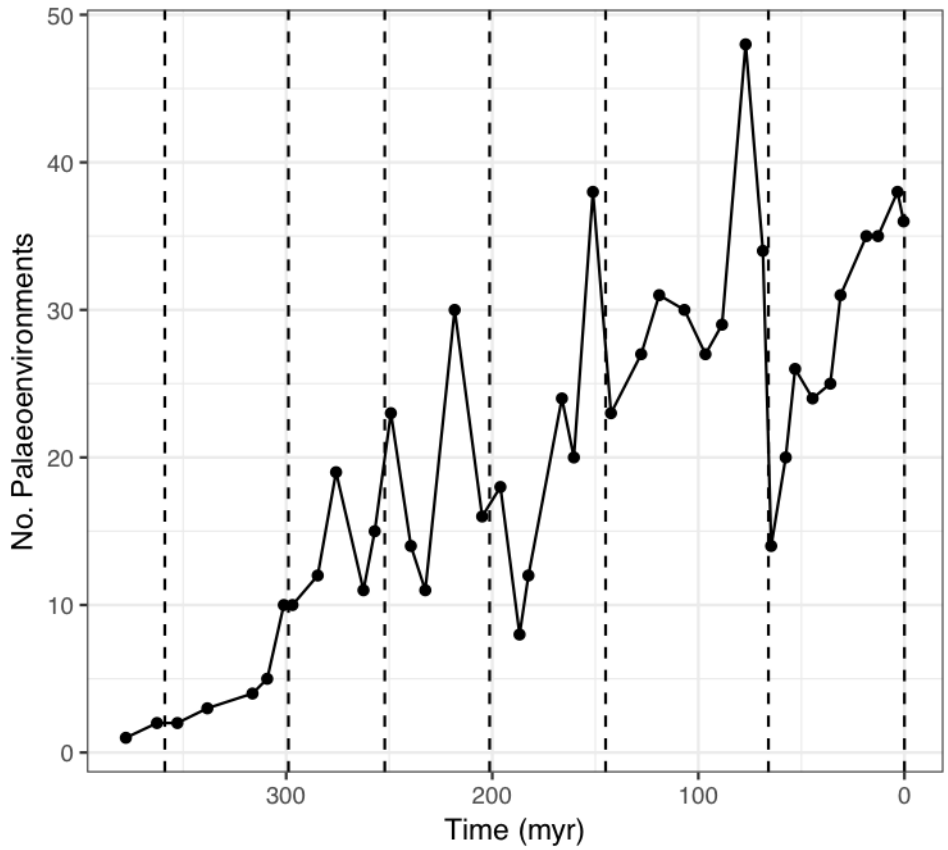

321

322 **Supplementary Fig. 12**. Proportions of lithified, unlithified and poorly lithified collections

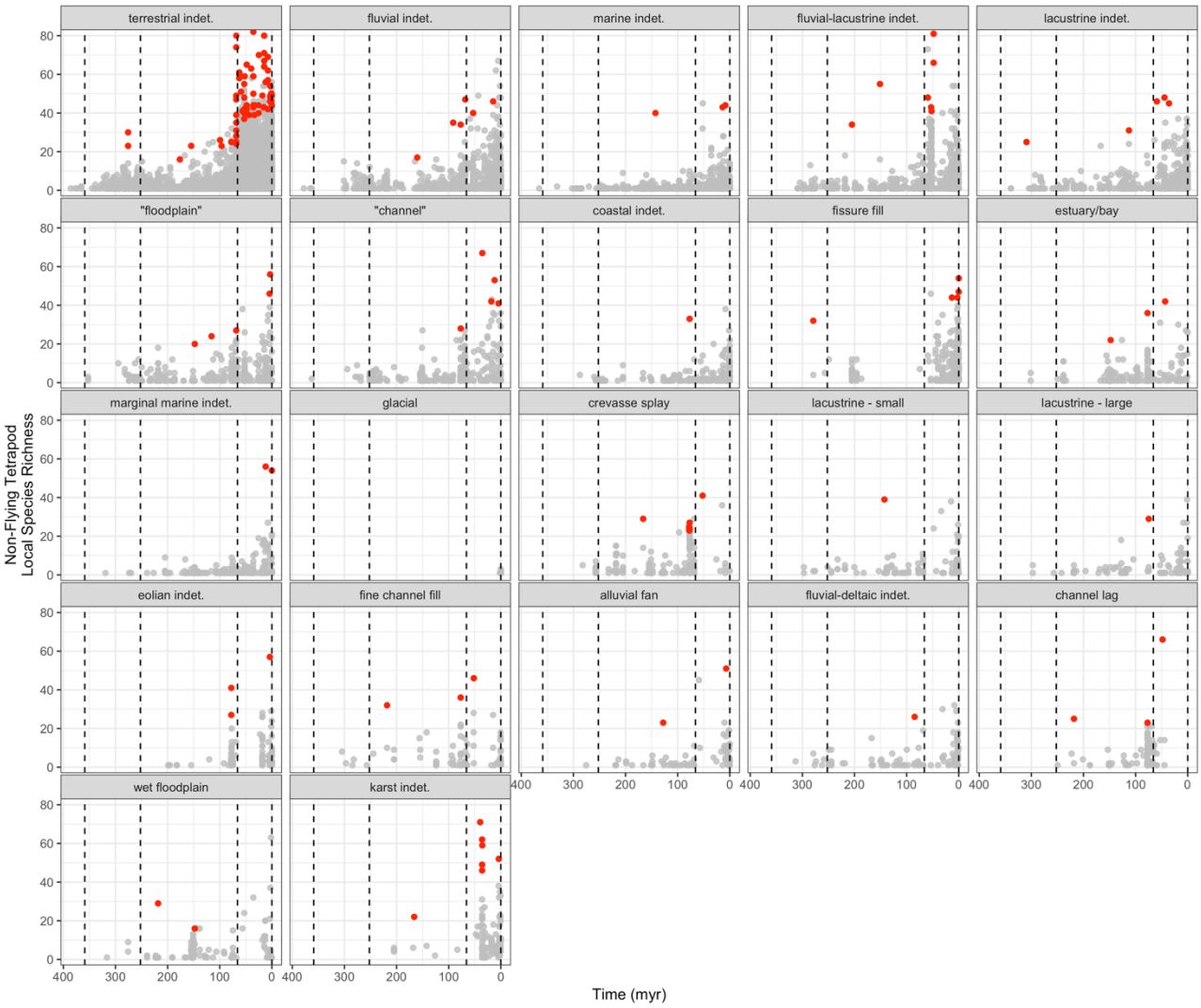
323 through time. **a**, Local richness for Phanerozoic non-marine tetrapods, colored by lithification

324 category. **b**, proportions of lithified, poorly-lithified and unlithified collections through time.

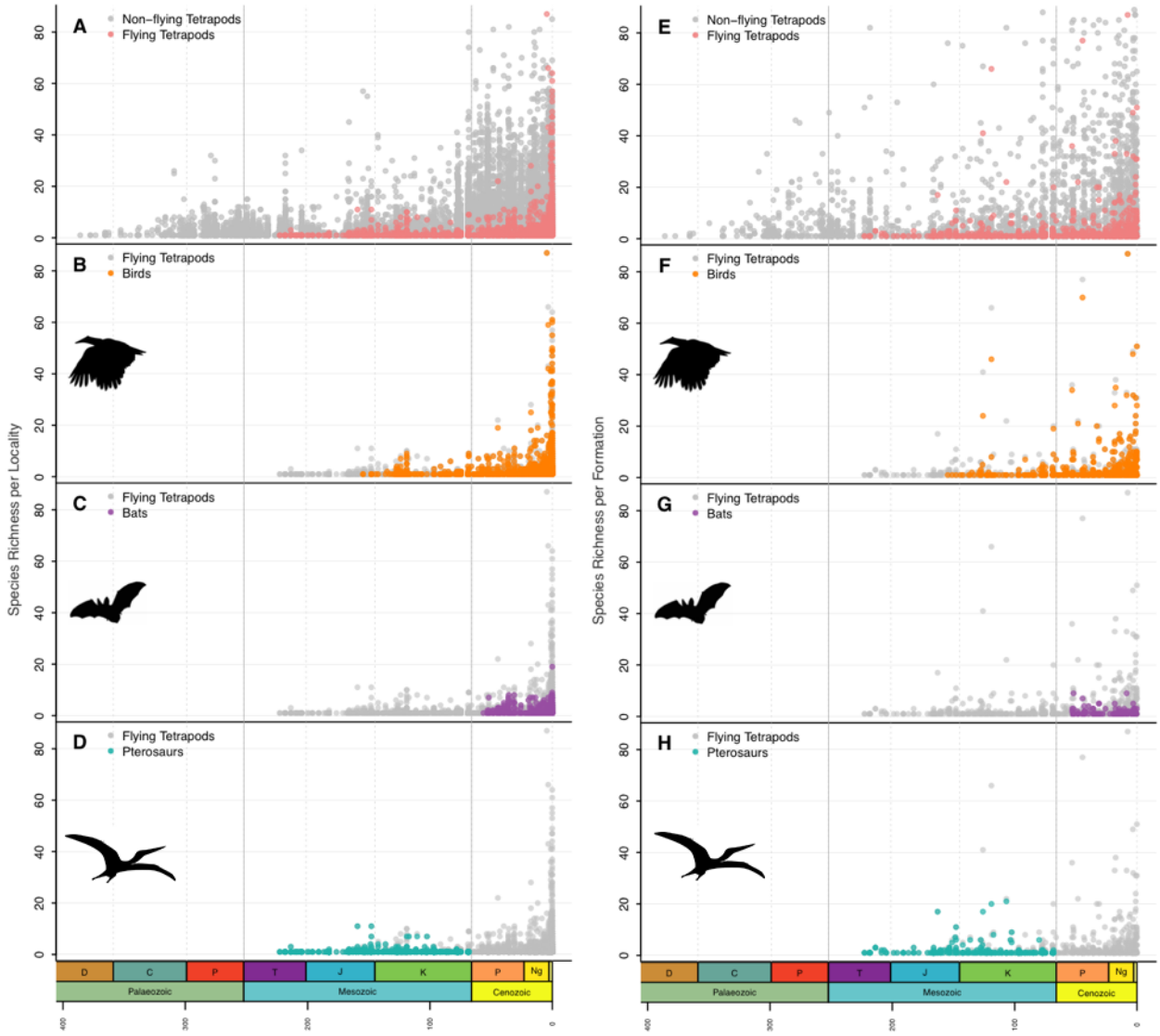
325




**Supplementary Fig. 13.** Counts of distinct environments (as recorded in the Paleobiology Database) per equal-length bin. The diversity of sampled environments within localities rose steadily through the Paleozoic–Mesozoic before falling across the K/Pg. Despite this, local richness rose abruptly in the Cenozoic, suggesting that sampled environments are not contributing to the broad Phanerozoic pattern of local richness.




**Supplementary Fig. 14.** Patterns of local richness within environmental types, as classified in the Paleobiology Database. Red points indicate exceptional localities (taxonomic co-occurrence restrictions = moderate, quantile cutoff = 0.99). Exceptional localities derive from a wide range of environment types, and exhibit no obvious temporal bias. Standardising for environmental type is difficult because most collections in the Paleobiology Database are recorded as “terrestrial indet.” due to limits of reporting in the palaeontological literature.




341

342 **Supplementary Fig. 15**. Local species richness for Phanerozoic flying tetrapods. Species per

343 collection for **a**, Flying tetrapods contrasted against non-flying tetrapods. **b**, Birds. **c**, Bats. **d**,

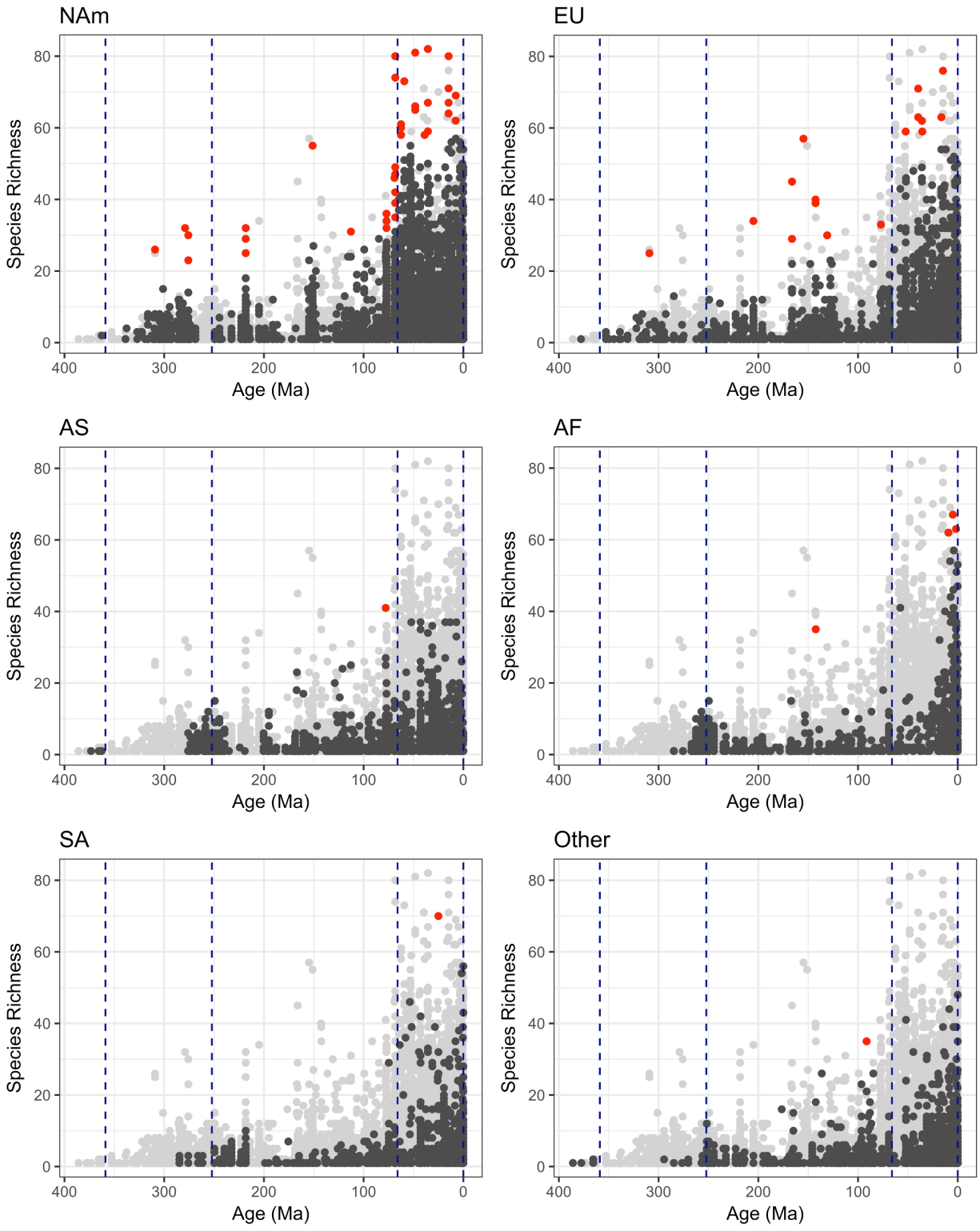
344 Pterosaurs. Species per formation for **e**, Flying tetrapods contrasted against non-flying

345 tetrapods. **f**, Birds. **g**, Bats. **h**, Pterosaurs.

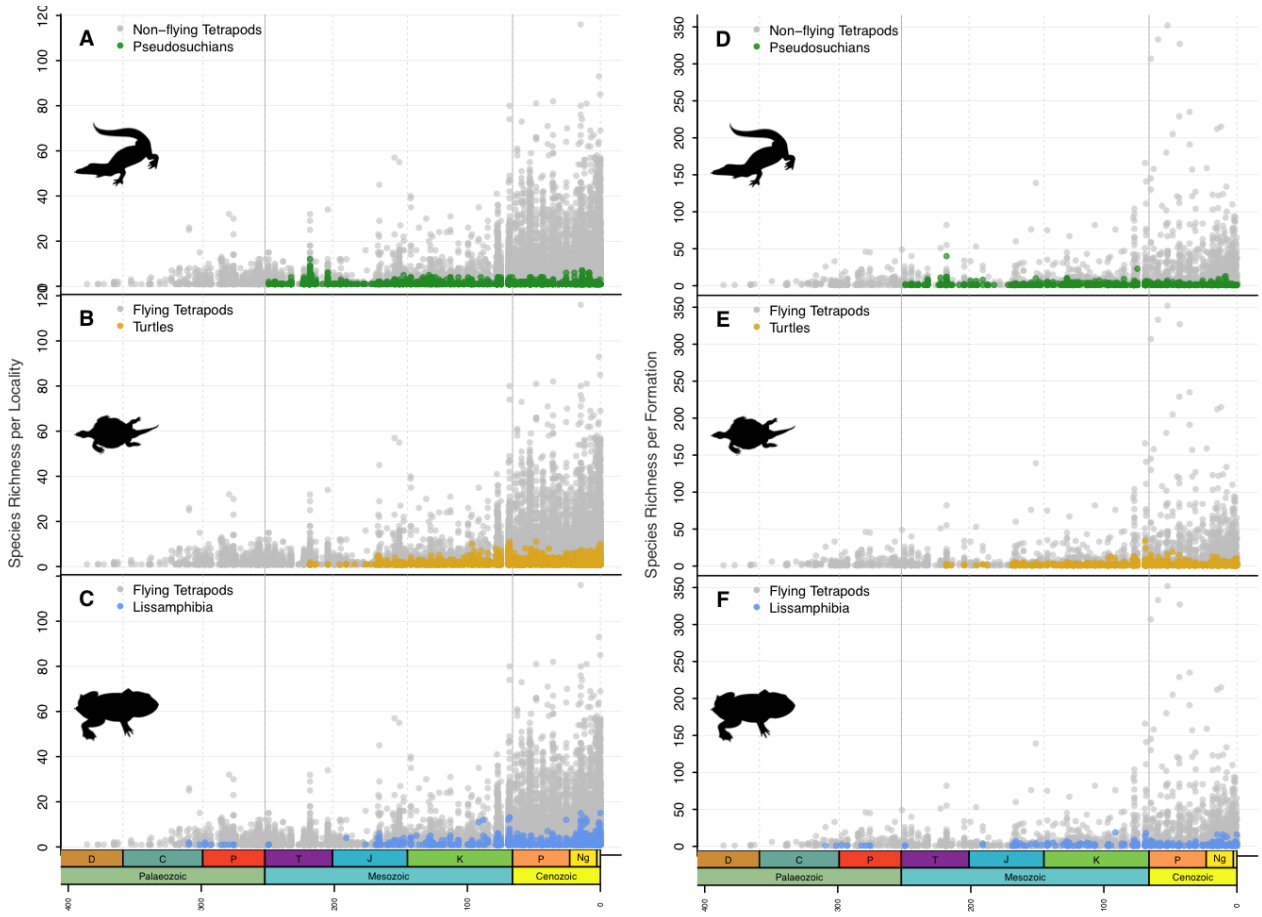
346




**Supplementary Fig. 16.** Paleolatitudes of collections through time, highlighting dominance of paleotemperate records and substantial increases in geographic coverage through time. Color range represents non-flying terrestrial tetrapod local richness. Black circles indicate exceptional localities.




**Supplementary Fig. 17**. Patterns of local richness in Phanerozoic non-flying tetrapods for key continental regions defined in the Methods section.




357

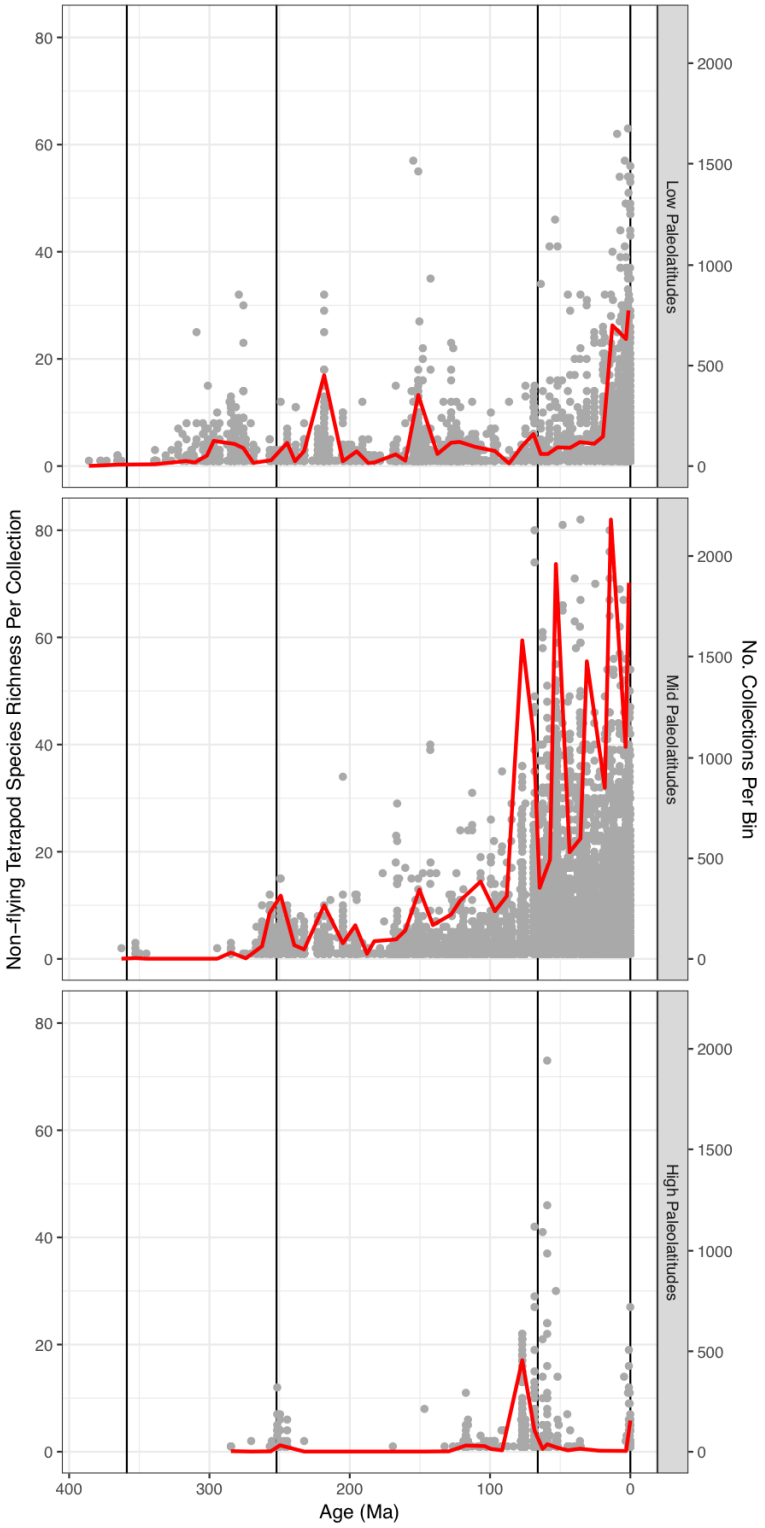
358 **Supplementary Fig. 18**. **Local species richness for less diverse tetrapod groups,**

359 **superimposed against the pattern for all non-flying tetrapods.** Species per collection for

360 **a**, Pseudosuchians. **b**, Turtles. **c**, Lissamphibians. Species per formation for **d**,

361 Pseudosuchians. **e**, Turtles. **f**, Lissamphibians.

362




363

364 **Supplementary Fig. 19**. Patterns of local richness within paleolatitudinal belts (low = 0–30°,

365 mid = 30–60°, high = 60–90°). Red line indicates number of collections per equal-length bin.

366

# Supplementary Tables

| Superlocality Set | Model | Adj. R-Squared | p-value | df | AIC | Akaike Weights | p-value (Time) | coef. (Time) | p-value (Phase | coef. (Phase 3) | p-value (Phase | coef. (Phase 4) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **None 0.99** | **~ Time +** | 0.689 | < 0.001 | 4 | 44.509 | 0.541 | < 0.001 | -0.002 | < 0.001 | -0.452 | < 0.001 | -0.744 |
| | **~ Time *** | 0.691 | < 0.001 | 6 | 44.875 | 0.408 | < 0.001 | -0.001 | < 0.001 | -0.329 | 0.002 | -1.188 |
| | **~ Phase** | 0.681 | < 0.001 | 3 | 49.322 | 0.051 | - | - | < 0.001 | -0.566 | < 0.001 | -1.096 |
| | **~ Time** | 0.641 | < 0.001 | 2 | 140.781 | 0 | < 0.001 | -0.005 | - | - | - | - |
| | **~ 1** | 0 | - | 1 | 380.663 | 0 | - | - | - | - | - | - |
| | **Pg0-Ng4** | -0.036 | 0.9 | 2 | - | - | - | - | - | - | - | - |
| | **Tr4-K8** | 0.219 | < 0.001 | 2 | - | - | - | - | - | - | - | - |
| | **C4-Tr5** | 0.002 | 0.27 | 2 | - | - | - | - | - | - | - | - |
| **None 0.995** | **~ Phase** | 0.679 | < 0.001 | 3 | 5.664 | 0.645 | - | - | < 0.001 | -0.509 | < 0.001 | -0.995 |
| | **~ Time +** | 0.679 | < 0.001 | 4 | 6.959 | 0.314 | < 0.001 | -0.001 | < 0.001 | -0.466 | < 0.001 | -0.851 |
| | **~ Time *** | 0.674 | < 0.001 | 6 | 10.625 | 0.041 | < 0.001 | -0.001 | < 0.001 | -0.5 | 0.21 | -0.625 |
| | **~ Time** | 0.624 | < 0.001 | 2 | 95.412 | 0 | < 0.001 | -0.005 | - | - | - | - |
| | **~ 1** | 0 | - | 1 | 209.887 | 0 | - | - | - | - | - | - |
| | **Pg0-Ng4** | -0.052 | 0.59 | 2 | - | - | - | - | - | - | - | - |
| | **Tr4-K8** | 0.114 | 0.012 | 2 | - | - | - | - | - | - | - | - |
| | **C4-Tr5** | -0.006 | 0.42 | 2 | - | - | - | - | - | - | - | - |
| **None 0.998** | **~ Phase** | 0.766 | < 0.001 | 3 | -20.037 | 0.705 | - | - | < 0.001 | -0.312 | < 0.001 | -0.944 |
| | **~ Time +** | 0.762 | < 0.001 | 4 | -18.416 | 0.255 | < 0.001 | 0.001 | < 0.001 | -0.347 | < 0.001 | -1.064 |
| | **~ Time *** | 0.759 | < 0.001 | 6 | -15.949 | 0.041 | < 0.001 | 0 | 0.055 | -0.293 | 0.003 | -1.568 |
| | **~ Time** | 0.673 | < 0.001 | 2 | 59.059 | 0 | < 0.001 | -0.006 | - | - | - | - |
| | **~ 1** | 0 | - | 1 | 110.663 | 0 | - | - | - | - | - | - |
| | **Pg0-Ng4** | -0.021 | 0.39 | 2 | - | - | - | - | - | - | - | - |
| | **Tr4-K8** | 0.412 | 0.002 | 2 | - | - | - | - | - | - | - | - |
| | **C4-Tr5** | -0.044 | 0.89 | 2 | - | - | - | - | - | - | - | - |
| **Moderate 0.99** | **~ Time +** | 0.54 | < 0.001 | 4 | 28.74 | 0.712 | < 0.001 | -0.002 | < 0.001 | -0.43 | 0.33 | -0.185 |
| | **~ Time *** | 0.539 | < 0.001 | 6 | 31.176 | 0.181 | < 0.001 | 0 | 0.004 | -0.325 | 0.72 | -0.204 |
| | **~ Phase** | 0.526 | < 0.001 | 3 | 32.647 | 0.107 | - | - | < 0.001 | -0.558 | < 0.001 | -0.59 |
| | **~ Time** | 0.343 | < 0.001 | 2 | 82.34 | 0 | < 0.001 | -0.003 | - | - | - | - |
| | **~ 1** | 0 | - | 1 | 146.992 | 0 | - | - | - | - | - | - |
| | **Pg0-Ng4** | 0.113 | 0.2 | 2 | - | - | - | - | - | - | - | - |
| | **Tr4-K8** | 0.019 | 0.14 | 2 | - | - | - | - | - | - | - | - |
| | **C4-Tr5** | -0.01 | 0.68 | 2 | - | - | - | - | - | - | - | - |
| **Moderate 0.995** | **~ Phase** | 0.598 | < 0.001 | 3 | -6.119 | 0.718 | - | - | < 0.001 | -0.487 | < 0.001 | -0.731 |
| | **~ Time +** | 0.593 | < 0.001 | 4 | -4.226 | 0.251 | < 0.001 | 0 | < 0.001 | -0.469 | 0.002 | -0.67 |
| | **~ Time *** | 0.585 | < 0.001 | 6 | -0.647 | 0.031 | < 0.001 | -0.001 | < 0.001 | -0.519 | 0.46 | -0.415 |
| | **~ Time** | 0.4 | < 0.001 | 2 | 25.755 | 0 | < 0.001 | -0.003 | - | - | - | - |
| | **~ 1** | 0 | - | 1 | 66.682 | 0 | - | - | - | - | - | - |
| | **Pg0-Ng4** | 0.001 | 0.35 | 2 | - | - | - | - | - | - | - | - |

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | **Tr4-K8** | -0.004 | 0.36 | 2 | - | - | - | - | - | - | - | - |
| | **C4-Tr5** | -0.019 | 0.63 | 2 | - | - | - | - | - | - | - | - |
| **Moderate 0.998** | **~ Phase** | 0.737 | < 0.001 | 3 | -20.991 | 0.684 | - | - | < 0.001 | -0.332 | < 0.001 | -0.825 |
| | **~ Time +** | 0.734 | < 0.001 | 4 | -19.716 | 0.276 | < 0.001 | -0.001 | 0.002 | -0.283 | 0.009 | -0.65 |
| | **~ Time *** | 0.722 | < 0.001 | 6 | -16.474 | 0.024 | < 0.001 | -0.002 | 0.018 | -0.384 | 0.29 | -0.616 |
| | **~ Time** | 0.662 | < 0.001 | 2 | -12.951 | 0.015 | < 0.001 | -0.003 | - | - | - | - |
| | **~ 1** | 0 | - | 1 | 25.187 | 0 | - | - | - | - | - | - |
| | **Pg0-Ng4** | 0.015 | 0.38 | 2 | - | - | - | - | - | - | - | - |
| | **Tr4-K8** | 0.154 | 0.082 | 2 | - | - | - | - | - | - | - | - |
| | **C4-Tr5** | 0.092 | 0.12 | 2 | - | - | - | - | - | - | - | - |
| **Stringent 0.99** | **~ Time +** | 0.519 | < 0.001 | 4 | 40.837 | 0.544 | < 0.001 | -0.001 | < 0.001 | -0.467 | 0.2 | -0.29 |
| | **~ Phase** | 0.511 | < 0.001 | 3 | 41.737 | 0.373 | - | - | < 0.001 | -0.574 | < 0.001 | -0.629 |
| | **~ Time *** | 0.513 | < 0.001 | 6 | 44.182 | 0.083 | < 0.001 | 0 | 0.003 | -0.393 | 0.71 | -0.237 |
| | **~ Time** | 0.31 | < 0.001 | 2 | 80.968 | 0 | < 0.001 | -0.003 | - | - | - | - |
| | **~ 1** | 0 | - | 1 | 123.415 | 0 | - | - | - | - | - | - |
| | **Pg0-Ng4** | 0.113 | 0.2 | 2 | - | - | - | - | - | - | - | - |
| | **Tr4-K8** | 0.003 | 0.28 | 2 | - | - | - | - | - | - | - | - |
| | **C4-Tr5** | -0.019 | 0.88 | 2 | - | - | - | - | - | - | - | - |
| **Stringent 0.995** | **~ Phase** | 0.579 | < 0.001 | 3 | 0.624 | 0.716 | - | - | < 0.001 | -0.475 | < 0.001 | -0.74 |
| | **~ Time +** | 0.574 | < 0.001 | 4 | 2.438 | 0.255 | < 0.001 | 0 | < 0.001 | -0.449 | 0.006 | -0.653 |
| | **~ Time *** | 0.563 | < 0.001 | 6 | 6.055 | 0.029 | < 0.001 | -0.001 | < 0.001 | -0.504 | 0.46 | -0.432 |
| | **~ Time** | 0.407 | < 0.001 | 2 | 23.743 | 0 | < 0.001 | -0.003 | - | - | - | - |
| | **~ 1** | 0 | - | 1 | 59.309 | 0 | - | - | - | - | - | - |
| | **Pg0-Ng4** | 0.001 | 0.35 | 2 | - | - | - | - | - | - | - | - |
| | **Tr4-K8** | 0.005 | 0.29 | 2 | - | - | - | - | - | - | - | - |
| | **C4-Tr5** | -0.021 | 0.54 | 2 | - | - | - | - | - | - | - | - |
| **Stringent 0.998** | **~ Phase** | 0.739 | < 0.001 | 3 | -15.451 | 0.74 | - | - | < 0.001 | -0.355 | < 0.001 | -0.848 |
| | **~ Time +** | 0.732 | < 0.001 | 4 | -13.753 | 0.232 | < 0.001 | -0.001 | 0.003 | -0.319 | 0.009 | -0.725 |
| | **~ Time** | 0.653 | < 0.001 | 2 | -7.259 | 0.015 | < 0.001 | -0.003 | - | - | - | - |
| | **~ Time *** | 0.713 | < 0.001 | 6 | -9.951 | 0.014 | < 0.001 | -0.001 | 0.038 | -0.367 | 0.33 | -0.599 |
| | **~ 1** | 0 | - | 1 | 25.647 | 0 | - | - | - | - | - | - |
| | **Pg0-Ng4** | 0.015 | 0.38 | 2 | - | - | - | - | - | - | - | - |
| | **Tr4-K8** | 0.154 | 0.082 | 2 | - | - | - | - | - | - | - | - |
| | **C4-Tr5** | -0.048 | 0.53 | 2 | - | - | - | - | - | - | - | - |

**Supplementary Table 1**. Relationships between species richness for Phanerozoic non-flying tetrapods within each of nine numerically-defined sets of exceptional localities ('Superlocality Set'), and combinations of time and diversification phase (see Methods). Bottom three rows in each set of exceptional localities show models for regressions within individual diversification phases.

| Bin | Stages | LAD | FAD | Midpoint | Duration |
|---|---|---|---|---|---|
| Ng4 | Calabrian, Middle P | 0 | 1.806 | 0.903 | 1.806 |
| Ng3 | Messinian, Zanclear | 1.806 | 7.246 | 4.526 | 5.44 |
| Ng2 | Langhian, Serravall | 7.246 | 15.97 | 11.608 | 8.724 |
| Ng1 | Aquitanian, Burdiga | 15.97 | 23.03 | 19.5 | 7.06 |
| Pg5 | Chattian, Rupelian | 23.03 | 33.9 | 28.465 | 10.87 |
| Pg4 | Bartonian, Priaboni | 33.9 | 41.3 | 37.6 | 7.4 |
| Pg3 | Lutetian | 41.3 | 47.8 | 44.55 | 6.5 |
| Pg2 | Ypresian | 47.8 | 56 | 51.9 | 8.2 |
| Pg1 | Selandian, Thanetia | 56 | 61.6 | 58.8 | 5.6 |
| Pg0 | Danian | 61.6 | 66 | 63.8 | 4.4 |
| K8 | Maastrichtian | 66 | 72.1 | 69.05 | 6.1 |
| K7 | Campanian | 72.1 | 83.6 | 77.85 | 11.5 |
| K6 | Coniacian, Santonia | 83.6 | 93.9 | 88.75 | 10.3 |
| K5 | Cenomanian | 93.9 | 100.5 | 97.2 | 6.6 |
| K4 | Albian | 100.5 | 113 | 106.75 | 12.5 |
| K3 | Aptian | 113 | 125 | 119 | 12 |
| K2 | Barremian, Hauteri | 125 | 132.9 | 128.95 | 7.9 |
| K1 | Berriasian, Valangi | 132.9 | 145 | 138.95 | 12.1 |
| J6 | Kimmeridgian, Tith | 145 | 157.3 | 151.15 | 12.3 |
| J5 | Callovian, Oxfordia | 157.3 | 166.1 | 161.7 | 8.8 |
| J4 | Bajocian, Bathonia | 166.1 | 170.3 | 168.2 | 4.2 |
| J3 | Aalenian, Toarcian | 170.3 | 182.7 | 176.5 | 12.4 |
| J2 | Pliensbachian | 182.7 | 190.8 | 186.75 | 8.1 |
| J1 | Hettangian, Sinemu | 190.8 | 201.3 | 196.05 | 10.5 |
| Tr5 | Rhaetian | 201.3 | 208.5 | 204.9 | 7.2 |
| Tr4 | Norian | 208.5 | 228 | 218.25 | 19.5 |
| Tr3 | Carnian | 228 | 237 | 232.5 | 9 |
| Tr2 | Ladinian | 237 | 242 | 239.5 | 5 |
| Tr1 | Anisian, Olenekian, | 242 | 252.17 | 247.085 | 10.17 |
| P5 | Changhsingian, Wu | 252.17 | 259.9 | 256.035 | 7.73 |
| P4 | Capitanian, Wordia | 259.9 | 268.8 | 264.35 | 8.9 |
| P3 | Kungurian, Roadian | 268.8 | 279.3 | 274.05 | 10.5 |
| P2 | Artinskian | 279.3 | 290.1 | 284.7 | 10.8 |
| P1 | Asselian, Sakmarian | 290.1 | 298.9 | 294.5 | 8.8 |
| C5 | Gzhelian, Kasimovi | 298.9 | 307 | 302.95 | 8.1 |
| C4 | Moscovian | 307 | 315.2 | 311.1 | 8.2 |
| C3 | Bashkirian, Serpukh | 315.2 | 330.9 | 323.05 | 15.7 |
| C2 | Visean | 330.9 | 346.7 | 338.8 | 15.8 |
| C1 | Tournaisian | 346.7 | 358.9 | 352.8 | 12.2 |
| D5 | Famennian | 358.9 | 372.2 | 365.55 | 13.3 |
| D4 | Frasnian | 372.2 | 382.7 | 377.45 | 10.5 |

**Supplementary Table 2.** Definitions of approximately equal-length time bins.

| Coll No | Coll Name | Formation | Period | Stage | Country | Geographic Scale | Stratigraphic Scale | Specimen Count | Sampling Level | Body-Size Range | Bulk Sampling | Environment | Lithology |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **18284** | Norden Bridge Quarry | Valentine | Cenozoic | Barstovian | US | local area | group of beds | 4288 | extensive | small-large | no | channel | sandstone |
| **18442** | Valentine Railway Quarries | Valentine | Cenozoic | Barstovian | US | local area | group of beds | 2760 | extensive | small-large | no | terrestrial indet. | |
| **46955** | Hambach mine horizon 6C | Ville | Cenozoic | MN 5 | DE | local area | group of beds | 1900 | extensive | small-large | yes | paralic indet. | lignite |
| **16919** | Calf Creek | Cypress Hills | Cenozoic | Chadronian | CA | outcrop | group of beds | 2900 | extensive | small-large | no | terrestrial indet. | not reported |
| **12840** | Le Bretou | | Cenozoic | Bartonian | FR | local area | bed | - | very extensive | small-large | yes | karst indet. | phosphorite |
| **16207** | Davis Ranch | Wind River | Cenozoic | Bridgerian | US | outcrop | group of beds | 1490 | extensive | small-large | unknown | fluvial-lacustrine indet. | mudstone |
| **16228** | Grizzly Buttes | Bridger | Cenozoic | Bridgerian | US | local area | group of beds | - | extensive | small-large | no | terrestrial indet. | not reported |
| **14939** | Cochrane 2 | Paskapoo | Cenozoic | Ti1 | CA | outcrop | group of beds | 1700 | extensive | small-large | no | fluvial-lacustrine indet. | siltstone |
| **14519** | Bushy Tailed Blowout | Lance | Mesozoic | Lancian | US | local area | bed | - | extensive | small-medium | yes | terrestrial indet. | shale |
| **14549** | Flat Creek 5 | Hell Creek | Mesozoic | Lancian | US | outcrop | bed | 217 | very extensive | small-very large | yes | terrestrial indet. | sandstone |

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **14566** | Hatcher No. 2 (UCMP V-5815) | Lance | Mesozoic | Lancian | US | outcrop | bed | - | extensive | small-very large | yes | terrestrial indet. | sandstone |
| **14585** | Lull 2 Quarry (UCMP V-5620) | Lance | Mesozoic | Late Maastrichtian | US | outcrop | bed | - | extensive | small-very large | yes | terrestrial indet. | sandstone |
| **45726** | Ksar Met-Lili, Anoual | Ksar Metlili | Mesozoic | Berriasian | MA | outcrop | group of beds | - | extensive | small-very large | yes | deltaic indet. | limestone |
| **28401** | Sunnydown Farm Quarry | Lulworth | Mesozoic | Middle Berriasian | UK | outcrop | group of beds | 26 | extensive | small-medium | yes | lacustrine - small | limestone |
| **12816** | Quarry 9, Como Bluff (YPM) | Morrison | Mesozoic | Kimmeridgian-Tithonian | US | outcrop | bed | 3748 | very extensive | small-very large | yes | fluvial-lacustrine indet. | mudstone |
| **24563** | Guimarota | Alcobaça | Mesozoic | Kimmeridgian | PT | local area | group of beds | - | very extensive | small-very large | yes | lagoonal | coal |
| **39075** | Kirtlington 3p (Mammal Bed) | Forest Marble | Mesozoic | Late Bathonian | UK | outcrop | bed | 1590 | extensive | small-very large | yes | mire/swamp | marl |
| **24883** | Dashanpu Dinosaur Quarry | Xiashaximiao | Mesozoic | Bajocian - Callovian | CN | outcrop | bed | - | extensive | medium-very large | no | lacustrine indet. | sandstone |
| **39160** | Placerias Quarry, UCMP A269 | Chinle | Mesozoic | Norian | US | outcrop | group of beds | - | very extensive | small-large | yes | wet floodplain | mudstone |
| **86388** | Abrahamskraal, Prince Albert | Abrahamskraal | Palaeozoic | Cap | ZA | local area | group of beds | 33 | very extensive | medium | no | floodplain | sandstone |
| **28256** | Coffee Creek, 34 | Arroyo | Palaeozoic | Kun | US | local area | formation | 50 | very extensive | small-large | no | terrestrial indet. | claystone |

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **28267** | Craddock Bonebed, Brush Creek, 38 | Arroyo | Palaeozoic | Kun | US | local area | group of beds | 61 | very extensive | small-large | no | terrestrial indet. | claystone |
| **67862** | Richards Spur | Garber | Palaeozoic | Leo | US | local area | formation | 320 | very extensive | small-large | no | fissure fill | claystone |
| **84835** | Nyrany | Kladno | Palaeozoic | West D | CZ | local area | formation | 38 | very extensive | small-medium | yes | lacustrine indet. | coal |
| **85292** | Linton | | Palaeozoic | West D | US | outcrop | bed | 69 | very extensive | small-large | yes | mire/swamp | coal |

375 **Supplementary Table 3**. Key information about selected exceptional localities yielding Phanerozoic terrestrial tetrapods.